DD DigiDelight

DIGICRM

On-Premises Deployment Specification

Infrastructure sizing, high availability and security for a Zambian commercial bank

Prepared for	**[Bank name], Lusaka, Zambia**
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Document version	**1.0 — Draft for discussion**
Date	**28 September 2026**
Classification	**Confidential**

Sized for a 100-user phase 1. Figures are planning estimates based on the workload assumptions in Section 2 and will be validated during a pilot.

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1. Summary

We recommend starting with 100 named users on a small two-site footprint: 2 physical hosts in production and 1 at the DR site, which also runs UAT. Estimated hardware cost is **USD 70k–95k** landed in Zambia, or close to zero on the bank's existing virtualisation cluster.

Environment	VMs	vCPU	RAM	Disk
Production (primary DC)	8	48	248 GB	~5 TB
DR (secondary DC)	3	24	128 GB	~3.2 TB
UAT (on the DR host)	2	8	32 GB	~1 TB
Optional on-prem AI	1 GPU server	16	128 GB	1 TB

The sizing covers 100 named users, 40 concurrent at peak and 500k customer records, with 5 years of data at 20% growth. All data and AI inference stays on bank premises.

> The design grows by adding nodes, not by rebuilding. Up to about 300 users needs one more app VM; beyond that we move to clustered data services (see 4.5 Growth path).

2. Workload assumptions

Phase 1 starts with 100 named users, such as the SME and corporate banking teams plus a small contact-centre group. Please confirm or correct each figure. The sizing scales roughly in proportion to them.

Dimension	Year 1	Year 5 (+20%/yr)	Notes
Named CRM users	100	210	RMs, branch managers, contact centre, marketing
Peak concurrent users	40	85	Month-end and campaign days, 08:00–17:00 CAT
Peak API requests	6 req/s	13 req/s	Staff UI plus integrations; ~150 ms median target
Contacts (individuals)	500 k	1 M	Loaded from core banking; nightly delta sync
Accounts (SME / corporate)	30 k	60 k	
Opportunities / leads	20 k/yr	40 k/yr	Loans, cards, bancassurance, deposits
Activities (calls, meetings, notes, emails)	1 M/yr	2 M/yr	
Support tickets	50 k/yr	100 k/yr	Email, web help centre, WhatsApp
Outbound messages (SMS, email, WhatsApp)	1 M/month	2 M/month	Peak 100 k in one campaign day
Automation / workflow runs	200 k/month	400 k/month	Event and schedule triggers
Core-banking sync	500 k upserts/night	1 M upserts/night	Runs 22:00–05:00
Attachments	300 GB in year 1	+200 GB/yr	KYC docs, ticket files, campaign assets
Web and chat analytics events	10 M/yr	20 M/yr	Only if the public site and app are tracked

Availability target: 99.5% in business hours. Planned maintenance happens in a Sunday window.

3. Application architecture

DigiCRM ships as Docker images. The components below match what runs in our current platform build, with no cloud services needed.

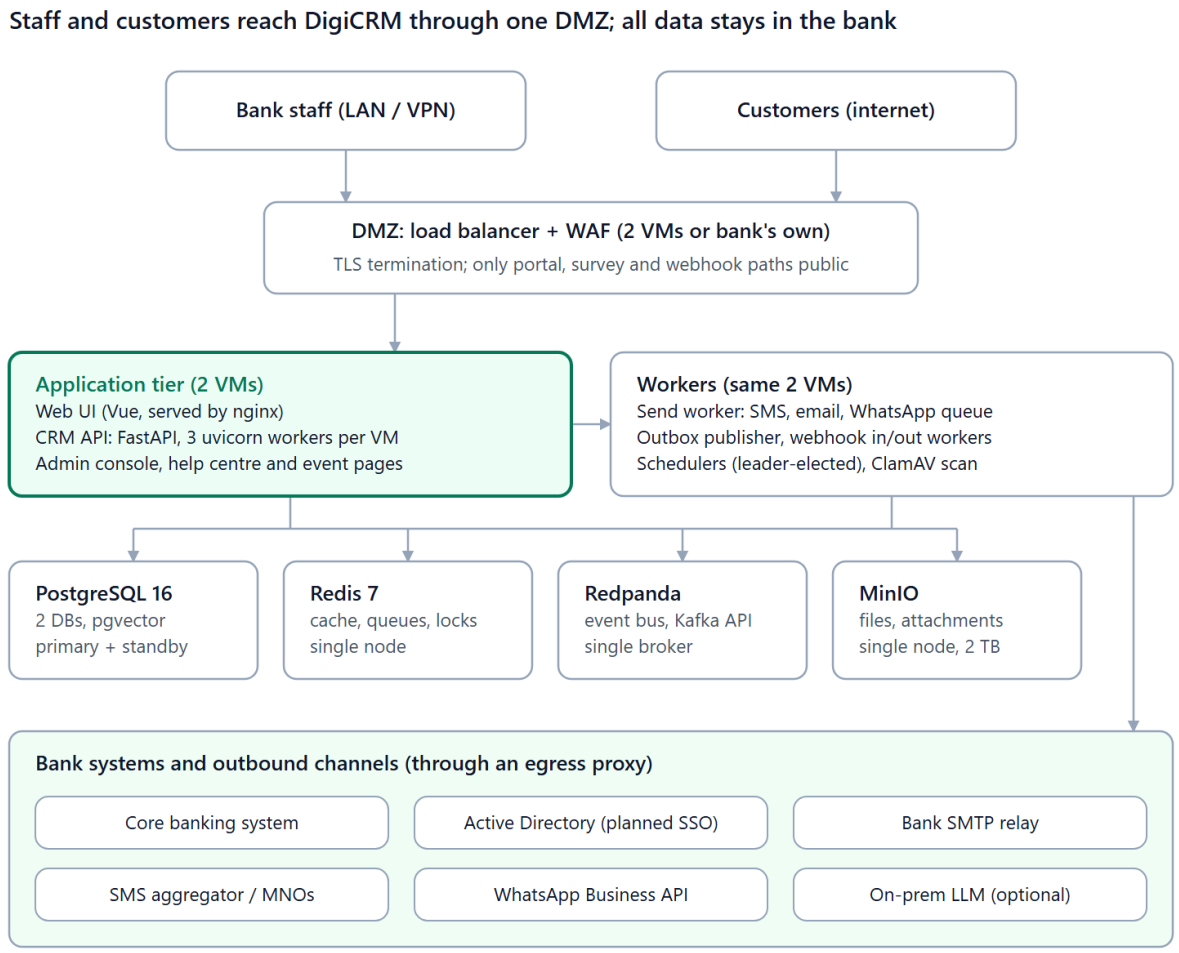


Figure 1 — DigiCRM on-premises architecture

The API and workers share the same four data stores. Only the workers and the Integration Hub call out of the bank, and they go through the bank's egress proxy.

Component	Role	Phase 1 (100 users)	When scaled up
Web UI	Vue single-page app served by nginx	On both app VMs	More app VMs
CRM API	FastAPI (Python 3.12). Each uvicorn worker uses ~0.8–1 GB RAM	3 workers per app VM	More workers, then more VMs
Schedulers	Campaign, workflow and sync crons. A Postgres advisory lock runs them on exactly one worker	Leader-elected, no extra VM	Same
Send worker, outbox publisher, webhook workers	Outbound SMS/email/WhatsApp queue; events to Redpanda; provider callbacks	On the app VMs	Dedicated worker VMs
PostgreSQL 16 + pgvector	Two databases: catalog (tenant, config) and business (CRM data)	Primary + streaming standby	Patroni cluster, 3 nodes + report replica

Component	Role	Phase 1 (100 users)	When scaled up
PgBouncer	Connection pooling in front of Postgres	On each database VM	Same
Redis 7	Cache, rate limits, send queue, locks	Single node, AOF persistence	Sentinel, 3 nodes
Redpanda	Event bus for automations and integrations	Single broker	3 brokers
MinIO	S3-compatible storage for attachments, KYC files, assets	Single node, RAID disks, mirrored to DR	4-node erasure set
ClamAV	Virus scan on every upload	On the app VMs	Same
Observability	Prometheus, Grafana, Loki, Alertmanager	1 VM	2 VMs

We deploy one tenant (the bank) on a single-tenant install. The SaaS admin console stays internal, for our support team and the bank's IT administrators.

4. Environments and server sizing

All 13 VMs fit on 3 physical hosts: 2 in production and 1 at DR. Alternatively they can run on the bank's existing VMware, Hyper-V or Proxmox cluster. OS: RHEL 9, Rocky Linux 9 or Ubuntu 24.04 LTS.

4.1 Production (primary DC)

Role	VMs	vCPU each	RAM each	Disk each	Runs
Load balancer / WAF	2	2	4 GB	50 GB	HAProxy + keepalived, or the bank's existing load balancer and WAF
App + workers	2	8	32 GB	200 GB	Web UI, CRM API (3 workers), send worker, outbox publisher, webhook workers, ClamAV
PostgreSQL	2	8	64 GB	1 TB SSD	Primary + streaming standby, PgBouncer, pgBackRest
Data services	1	8	32 GB	2 TB	Redis, Redpanda (single broker), MinIO (single node)
Monitoring + bastion	1	4	16 GB	500 GB	Prometheus, Grafana, Loki, Alertmanager, jump host, image registry
Total	**8**	**48**	**248 GB**	**~5 TB**	

One of each pair runs on each host, so either host can fail. The surviving host carries the full load: it needs ~150 GB of its 256 GB RAM.

4.2 DR (secondary DC)

DR runs warm: the database streams continuously and MinIO mirrors files. The app VM is kept stopped and started on failover.

Role	VMs	vCPU each	RAM each	Disk each
App + workers	1	8	32 GB	200 GB
PostgreSQL replica	1	8	64 GB	1 TB SSD
Data services	1	8	32 GB	2 TB
Total	**3**	**24**	**128 GB**	**~3.2 TB**

4.3 UAT / pre-production

UAT runs on the DR host. During a real DR event it is shut down to free capacity.

Role	VMs	vCPU	RAM	Disk
App + data services	1	4	16 GB	500 GB
PostgreSQL	1	4	16 GB	500 GB

> UAT holds masked production data. We never copy real customer PII into UAT without masking.

4.4 How the numbers were derived

- **API**: 40 concurrent users make ~6 req/s at peak. One uvicorn worker handles 8–12 req/s, so 6 workers give ~10× headroom. One app VM alone carries the full load.
- **Memory**: 3 API workers (~3 GB) plus send, outbox and webhook workers, ClamAV and headless Chromium use ~10 GB of the 32 GB.
- **Database**: the year-1 database (~30 GB) fits entirely in 64 GB RAM. Settings: shared_buffers 16 GB, effective_cache_size 48 GB, max_connections 200 behind PgBouncer.
- **Sends**: 100 k messages on a peak day are ~1 msg/s averaged, ~20 msg/s in a burst. The SMS gateway's throughput is the real limit.

4.5 Growth path

Growth	Change	Extra hardware (approx.)
Up to ~300 users	+1 app VM; host RAM to 384 GB	USD 5–10k
300–1,000 users	3rd production host; 3-node Redis, Redpanda and MinIO; 3rd database node	USD 60–100k
1,000+ users	Full clustered layout: dedicated worker, database and storage nodes	Quoted on request

Each step adds VMs to the same architecture. No data migration or re-platforming is needed.

5. Hardware and software cost estimate

Hardware for production, DR and UAT comes to about **USD 70k–95k** landed in Zambia. If DigiCRM runs on the bank's existing virtualisation cluster, new hardware drops to **USD 0–15k** (backup storage at most).

Item	Qty	Config	Estimate (USD)
Production hosts	2	1 × 32-core CPU, 256 GB RAM, 4 × 3.84 TB NVMe, 25 GbE	32–40k
DR host (also runs UAT)	1	Same model	16–20k
Backup storage (NAS, ~8 TB usable)	1		5–8k
Networking	—	Bank's existing switches and firewall	0–5k
Import, freight, reseller margin		+20–30% on hardware	15–20k
Optional AI: GPU server	1	1 × NVIDIA L40S 48 GB	+20–30k
Hardware total (landed, excl. AI)			**~70–95k**

5.1 Yearly software

- OS: Rocky Linux or Ubuntu is free; RHEL is about USD 800–1,300 per server per year.
- Hypervisor: Proxmox or Hyper-V is low or no cost. VMware for 3 hosts is roughly USD 10k–30k a year under current Broadcom licensing.
- DigiCRM's own stack (PostgreSQL, Redis, Redpanda, MinIO, monitoring) has no licence fees.

> These are list-price estimates for enterprise servers (Dell PowerEdge R760 / HPE DL380 Gen11 class) with 3-year next-business-day support, not supplier quotes. They exclude 16% VAT, which banks can often reclaim.

6. Storage and database sizing

The business database grows to about 170 GB by year 5, well inside the 1 TB SSD per database VM. Files reach about 1.1 TB, inside the 2 TB data-services disk.

Data	Row size (incl. indexes)	Year 1	Year 5 (cumulative)
Contacts + accounts + custom fields	~4 KB	2 GB	4 GB
Activities	~2 KB	2 GB	15 GB
Message send log + delivery events	~1 KB	12 GB	90 GB
Tickets + email threads	~10 KB	0.5 GB	4 GB
Audit log and record history	~0.5 KB	1.5 GB	11 GB
Analytics events (if enabled)	~0.5 KB	5 GB	37 GB
AI embeddings (pgvector, 1,536 dims)	~6 KB	3 GB	6 GB
Attachments (MinIO)		300 GB	~1.1 TB
Database total		**~26 GB**	**~170 GB**

6.1 Backup and retention

- **Database**: pgBackRest does a weekly full, daily differential and continuous WAL archiving. Point-in-time restore goes back 35 days. Backup store needs ~1 TB at year 5.
- **Files**: MinIO mirrors to DR, with versioning and 35-day retention of deleted objects.
- **Long-term**: a monthly full to the bank's tape or immutable backup (WORM) platform, kept for the bank's statutory period.
- **Restore test**: quarterly, restoring into UAT, with timings recorded against the RTO.

Retention of CRM records (activities, communications, KYC documents) should follow the bank's records policy. Zambian anti-money-laundering record keeping is commonly cited as 10 years; the bank's compliance team should confirm.

7. Network, security and compliance

The design keeps every customer record, file and AI prompt inside the bank's data centres in Zambia. That meets data-residency expectations without legal review of each cloud service.

7.1 Network zones

Zone	Contains	Inbound from	Outbound to
DMZ	Load balancers, WAF	Internet (443 only, public paths); bank LAN (443)	App tier (443)
Application	App and worker nodes	DMZ only	Data zone; egress proxy
Data	PostgreSQL, Redis, Redpanda, MinIO	Application zone only	DR site (replication)
Management	Bastion, registry, observability	Bank admin VLAN, with MFA	All zones (SSH, metrics)

Public paths are the help centre, survey and event pages, the chat widget and provider webhooks. All staff screens stay on the LAN or VPN, or the bank can keep the whole system internal.

7.2 Security controls

- TLS 1.2+ everywhere, with certificates from the bank's internal CA. Replication links between DCs are encrypted.
- Encryption at rest: LUKS or SAN encryption on database and MinIO volumes. Integration secrets are already encrypted in the application.
- Role-based access, record-level sharing and field security are built in. Audit logs go to the bank's SIEM via syslog or OpenTelemetry.
- Every upload is virus-scanned by ClamAV. Signatures update from an internal mirror.
- Container images are built by us, scanned (Trivy) and signed. They are then pushed to the bank's Harbor registry; production never pulls from the internet.
- Yearly penetration test by a bank-approved third party before go-live and after major releases.

7.3 Regulatory points to confirm with the bank

- **Data Protection Act No. 3 of 2021**: registration as data controller/processor, and limits on cross-border transfer. That matters for WhatsApp and any cloud AI.
- **Bank of Zambia cyber and information risk guidelines**, plus outsourcing rules: DigiCRM is a third-party system, so our support access needs a contract and audit rights.
- **Anti-money-laundering record keeping**: retention periods drive backup sizing.

> These references are from general knowledge, not a legal review. The bank's compliance office should confirm the current versions.

8. High availability, backup and DR

Targets: no data loss for a database or host failure, with a few minutes of disruption. Up to 5 minutes of data loss and 2 hours to recover for a full site loss.

Failure	Handled by	RPO	RTO
One app VM	Load balancer health checks; scheduler lock moves to the other VM	0	< 30 s
Database primary	Promote the streaming standby (scripted); PgBouncer repointed	0	< 5 min
Redis, Redpanda or MinIO VM	Hypervisor restarts the VM on the other host; Redis AOF and disk data survive	~1 s	< 10 min
One physical host	Surviving host runs all production VMs	0	< 10 min
Whole primary DC	Promote DR Postgres, switch DNS, start DR app VM	≤ 5 min	≤ 2 h (runbook)
Data corruption or bad release	pgBackRest point-in-time restore; image rollback from the registry	to the minute	2–4 h

- The DR site should be at least 20 km from the primary, on a separate power grid feed. Between them we need ≥ 100 Mbps and < 20 ms latency.
- Load-shedding is a real risk in Zambia. Each DC needs UPS for ≥ 30 minutes plus generator backup. Postgres, Redis and MinIO disks need power-loss protection.
- DR failover is rehearsed twice a year, with the result reported to the bank's risk committee.

9. AI, messaging and integrations

Every DigiCRM integration works through an allow-listed egress proxy. The only ones that must leave the country are WhatsApp and any cloud AI the bank chooses to enable.

Integration	On-prem option	Leaves the bank?	Notes
Email out	Bank's SMTP relay (Exchange or Postfix)	No	Outlook 365 connector available if the bank uses M365
Email in (support tickets)	IMAP/Exchange mailbox	No	Known-contact-only ticket creation is supported
SMS	Local aggregator or MNO SMPP/HTTP APIs (MTN, Airtel, Zamtel)	To the carrier	Throughput is set by the carrier contract
WhatsApp	Meta WhatsApp Business API via a BSP	Yes: Meta's cloud	Needs a data-transfer assessment under the DPA
Core banking	Integration Hub connectors, nightly batch plus real-time webhooks	No	Read-only service account; the bank's ESB if one exists
AI features	Self-hosted model on GPU servers via our OpenAI-compatible provider	No	Or disable AI entirely; everything else still works
Enrichment (Apollo, Clearbit)	None on-prem	Yes	Recommend leaving these off for a bank
Staff sign-in	DigiCRM local accounts (password only) today	No	Gap: no SAML/OIDC/AD login and no staff MFA yet. Both are needed before go-live

9.1 On-prem AI (optional)

DigiCRM's assistant, AI drafting, summaries and AI tracking planner can call any OpenAI-compatible endpoint. On-prem that means vLLM serving an open-weight model.

Item	Spec
GPU server	1 server with 1 × NVIDIA L40S 48 GB
CPU / RAM / disk	16 cores, 128 GB RAM, 1 TB NVMe
Model class	An 8–30B-parameter instruction model (quantised) for chat and drafting, plus a small embedding model for pgvector search
Capacity	About 10 concurrent AI requests with a few seconds of latency. Plenty for 100 users

> The bank should start with AI off. Turn it on per feature after a model-risk review.

10. Software prerequisites, monitoring and open questions

The stack is open source and needs no per-core licences, apart from the OS support the bank already buys.

Layer	Version	Licence
OS	RHEL 9 / Rocky 9 / Ubuntu 24.04 LTS	Bank's existing subscription
Container runtime	Docker Engine 27+ with Compose, or Kubernetes (RKE2 / OpenShift) if the bank already runs it	Open source
PostgreSQL	16 with pgvector; PG18 once certified	PostgreSQL licence
PgBouncer, pgBackRest (Patroni + etcd when scaled)	Current stable	Open source
Redis	7.x	See note below
Redpanda	Community edition	BSL; free for self-hosted use
MinIO	Current release	AGPL v3; commercial licence if the bank's legal team prefers
HAProxy / NGINX, ClamAV, Prometheus, Grafana, Loki, Tempo	Current stable	Open source

Redis 7.4 and later changed licence; we can pin 7.2 or use Valkey, a drop-in fork. Kubernetes is optional: our Compose files and Ansible are enough for this footprint, and they need fewer specialist skills to run.

10.1 Monitoring and operations

- Grafana dashboards and Alertmanager alerts cover API latency, error rate, queue depth, replication lag, disk and certificate expiry. Alerts go to the bank's NOC via email or SMS.
- Releases run blue/green on the app tier, with database migrations run first. Normal releases are monthly; security fixes within 72 hours.
- Support is a named L2/L3 team with remote access through the bank's PAM tool only, session-recorded.

10.2 Open questions for the bank

- ☐ Confirm the 100-user phase 1 scope, expected growth, and the customer and message volumes in the assumptions table
- ☐ Which core banking system and ESB, and is real-time or nightly sync enough?
- ☐ Run on the bank's existing virtualisation cluster, or buy the 3 dedicated hosts?
- ☐ Existing hypervisor, SAN and backup platform, and whether dedicated database hosts are allowed
- ☐ Staff login: Active Directory via SAML, OIDC or LDAP?
- ☐ Should any screen be reachable from the internet, or is the system LAN/VPN only?
- ☐ SMS provider and contracted throughput; is WhatsApp in scope?
- ☐ AI features in or out of phase 1, and is a GPU purchase approved?
- ☐ Required RPO/RTO and the DR site location
- ☐ Record-retention periods and data-residency rulings from compliance